

CE80362 : Circuit Analysis

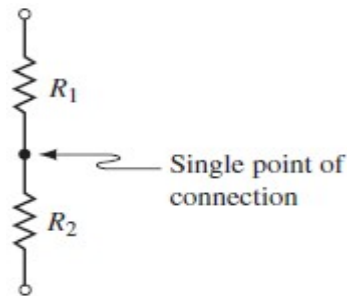
Lecture 2:

Basic Preliminaries on Circuit Analysis, Mesh (loop) and Nodal Analysis.

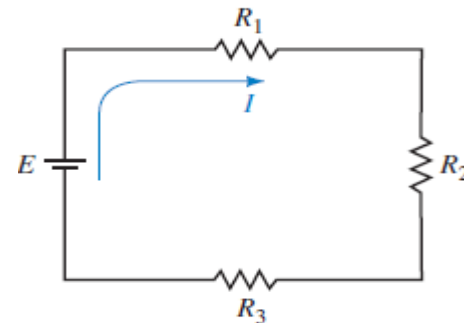
Basic Preliminaries on Circuit Analysis

- **Series Circuits**
- An electric circuit is the combination of any number of sources and loads connected in any manner that allows charge to flow.
- The electric circuit may be simple, such as a circuit consisting of a battery and a light bulb or complex. All electric circuits obtain their energy either from a direct current (dc) source or from an alternating current (ac) source.
- Although ac circuits have fundamental differences when compared with dc circuits, the laws, theorems, and rules that you learn in dc circuits apply directly to ac circuits as well.

- A series circuit is constructed by combining various elements in series.
 - *Two elements are said to be in series if they are connected at a single point and if there are no other current-carrying connections at this point.*
 - *The current is the same everywhere in a series circuit.*



Resistors in series.



Series circuit.

- **Kirchhoff's Voltage Law**

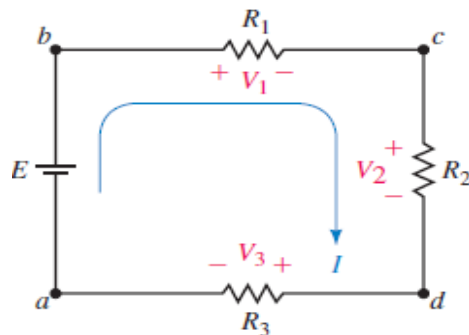
The *most important laws* of electricity is Kirchhoff's voltage law (KVL), which states the following:

- *The summation of voltage rises and voltage drops around a closed loop is equal to zero.*

Symbolically, this may be stated as follows:

- $\sum V = 0$ *for a closed loop*
- **A closed loop** is defined as any path that originates at a point, travels around a circuit, and returns to the original point without retracing any segments.
- When there are loads, an alternate Kirchhoff's voltage law is applied:
 - *The summation of voltage rises is equal to the summation of voltage drops around a closed loop.*
 - $\sum E_{Total} = \sum V_{drop}$

- Example 1

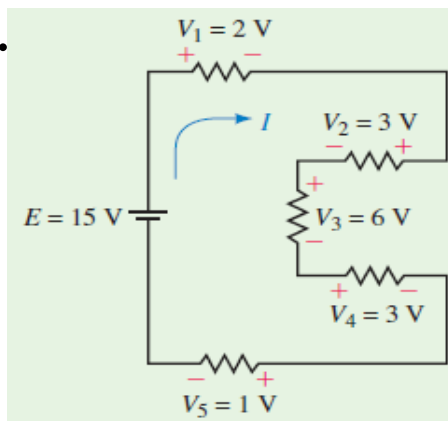


$$E = V_1 + V_2 + V_3$$

The voltage dropped across the resistors react in opposite direction compared to the direction of the source.

- Example 2: Verify Kirchhoff's voltage law for the following circuits.

(a)

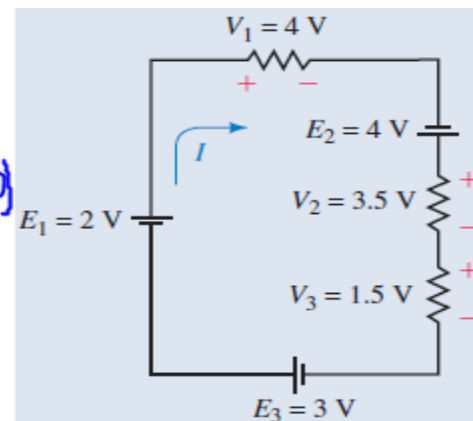


Answer (a): If we follow the direction of the current, we write the loop equation as

$$E = V_1 + V_2 + V_3 + V_4 + V_5$$

$$E - V_1 - V_2 - V_3 - V_4 - V_5 = 15 - 2 - 3 - 6 - 3 - 1 = 0$$

(b)



Answer (b): If we follow the direction of the current, we write the loop equation as

$$E_1 + E_2 + E_3 = V_1 + V_2 + V_3$$

$$E_1 + E_2 + E_3 - V_1 - V_2 - V_3 = 2 + 4 + 3 - 4 - 3.5 - 1.5 = 0$$

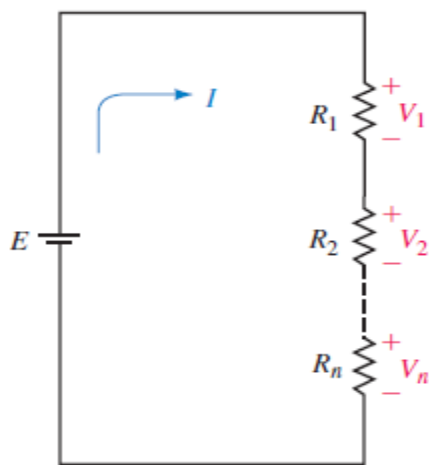
Resistors in Series

Since the circuit is a closed loop, *the voltage source will cause a current I in the circuit*. This current in turn produces a voltage drop across each resistor, where

$$\bullet V_x = IR_x$$

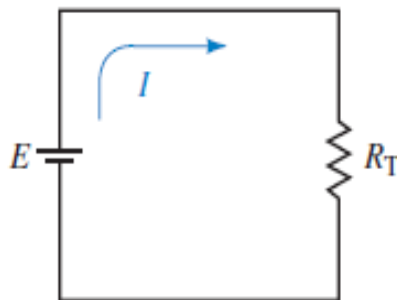
Applying Kirchhoff's voltage law to the closed loop gives:

$$\begin{aligned} E &= V_1 + V_2 + \dots + V_n \\ &= IR_1 + IR_2 + \dots + IR_n \\ &= I(R_1 + R_2 + \dots + R_n) \end{aligned}$$



(a)

If we were to replace all the resistors of fig. (a) with a total equivalent resistance, R_T , then the circuit would appear as in fig.(b)

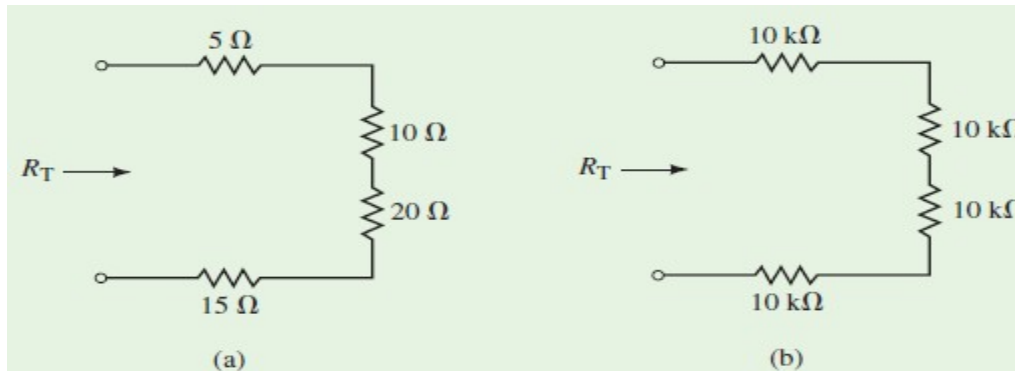


(b)

Hence in series network,

$$R_T = \sum R_i, \quad i = 1, 2, \dots, n$$

- Examples: Determine the total resistance for each of the following networks



Solution

- $R_T = 5 + 10 + 20 + 15 = 50 \Omega$
- $R_T = 4(10 \text{ k}\Omega) = 40 \text{ k}\Omega$

Any voltage source connected to the terminals of a network of series resistors will provide the same current as if a single resistance, having a value of R_T , **hence $I = \frac{E}{R_T}$ (Ohm's law)**

- The power dissipated** by each resistor is:

- Hence, $P_T = EI$, with $E = V_1 + V_2 + \dots + V_n$

- $P_T = P_1 + P_2 + \dots + P_n$

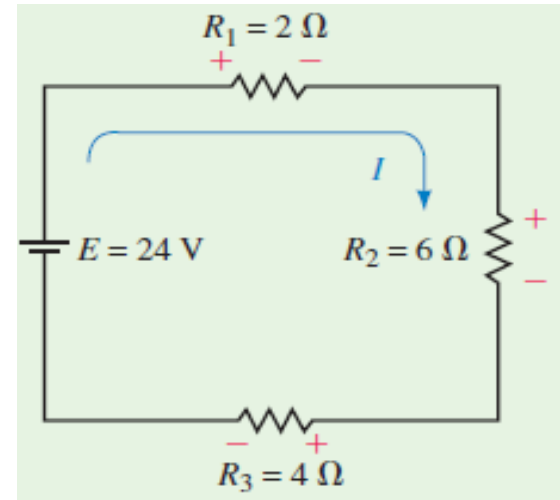
Meaning that **energy must be conserved, the power delivered by the voltage source is equal to the total power dissipated by all the resistors.**

$$\begin{aligned}
 P_1 &= V_1 I = \frac{V_1^2}{R_1} = I^2 R_1 \quad [\text{watts, W}] \\
 P_2 &= V_2 I = \frac{V_2^2}{R_2} = I^2 R_2 \quad [\text{watts, W}] \\
 &\vdots \\
 P_n &= V_n I = \frac{V_n^2}{R_n} = I^2 R_n \quad [\text{watts, W}]
 \end{aligned}$$

• Example for Enhancement (1) :

(a) *For the series circuit shown in Fig. (a), find the following quantities:*

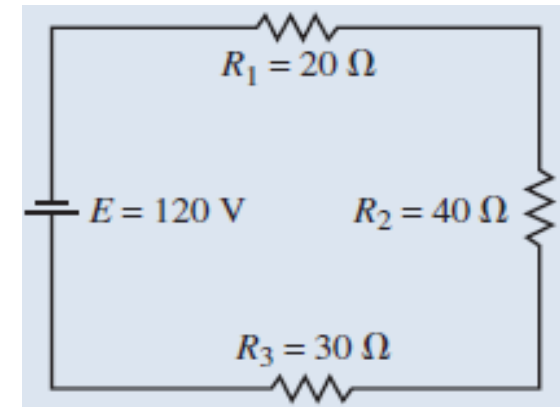
- Total resistance, R_T
- Circuit current, I .
- Voltage across each resistor
- Power dissipated by each resistor.
- Power delivered to the circuit by the voltage source.
- Verify that the power dissipated by the resistors is equal to the power delivered to the circuit by the voltage source.



(a)

(b) *For the series circuit shown in Figure (b), find the following quantities:*

- Total resistance, R_T .
- The direction and magnitude of the current, I .
- Polarity and magnitude of the voltage across each resistor.
- Power dissipated by each resistor.
- Power delivered to the circuit by the voltage source.
- Show that the power dissipated is equal to the power delivered.



(b)

The Voltage Divider Rule

- The voltage divider rule **allows us to determine the voltage across any series resistance in a single step**, without first calculating the current.
- **Statement:** *“The voltage dropped across any series resistor is proportional to the magnitude of the resistor. The total voltage dropped across all resistors must equal the applied voltage source(s) by Kirchhoff’s Voltage Law-KVL.”*
- This is done by applying voltage divider rule on any of the resistors as a simple equation:

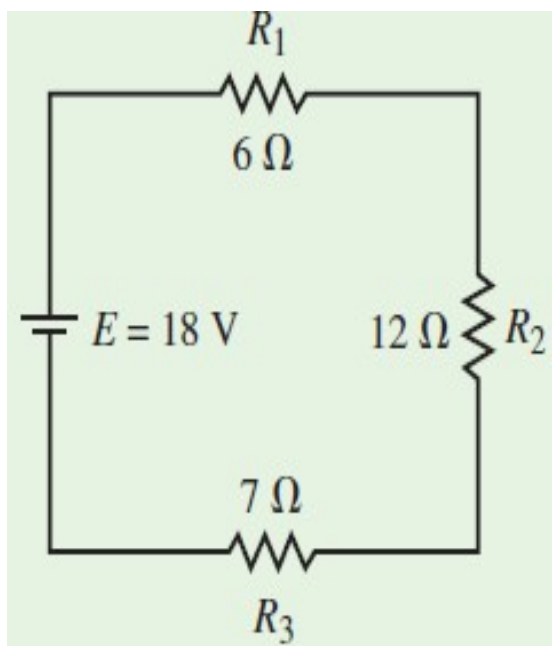
$$V_x = \frac{R_x E}{R_1 + R_2 + R_3 + \dots + R_n}$$

where V_x is a Voltage across the concerned resistor R_x in series.

- **Generally, for any number of resistors the voltage drop across any resistor may be found as:**

$$V_x = \frac{R_x}{R_T} E$$

- Example (1):** Use the voltage divider rule to determine the voltage across each of the resistors in the circuit shown below. Show that the summation of voltage drops is equal to the applied voltage rise in the circuit.



Solution

$$R_T = 6\ \Omega + 12\ \Omega + 7\ \Omega = 25.0\ \Omega$$

$$V_1 = \left(\frac{6\ \Omega}{25\ \Omega} \right) (18\text{ V}) = 4.32\text{ V}$$

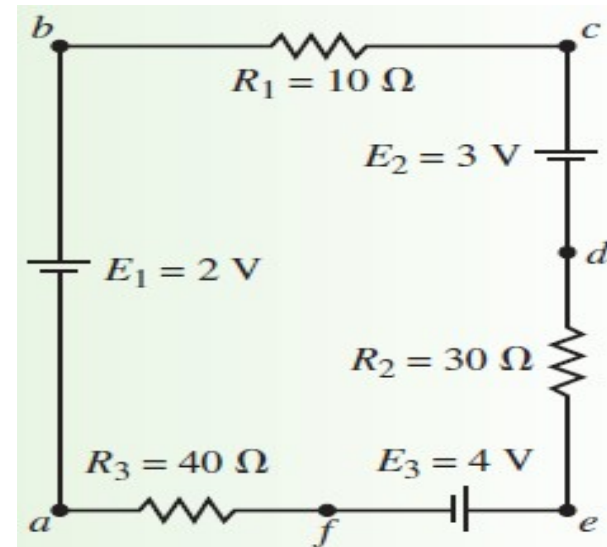
$$V_2 = \left(\frac{12\ \Omega}{25\ \Omega} \right) (18\text{ V}) = 8.64\text{ V}$$

$$V_3 = \left(\frac{7\ \Omega}{25\ \Omega} \right) (18\text{ V}) = 5.04\text{ V}$$

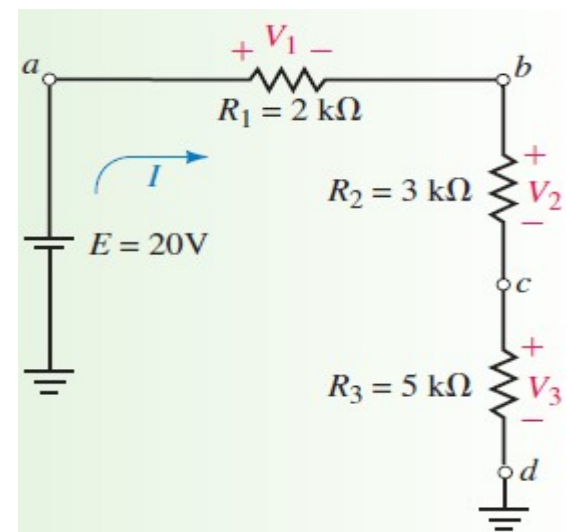
The total voltage drop is the summation:

$$V_T = V_1 + V_2 + V_3 = 4.32 + 8.64 + 5.04 = 18\text{ V} \\ = E$$

- **Example for Enhancement (2) :**
(1) Apply the Voltage divider rule to determine the following voltages: V_{ac} , V_{ad} , V_{cf} , and V_{eb} .



- (2)** Applying the voltage divider rule to determine the voltages V_a , V_b , V_c , and V_d .



V_{cf} , and V_{eb}

find the voltages V_{ac} , V_{ad} ,

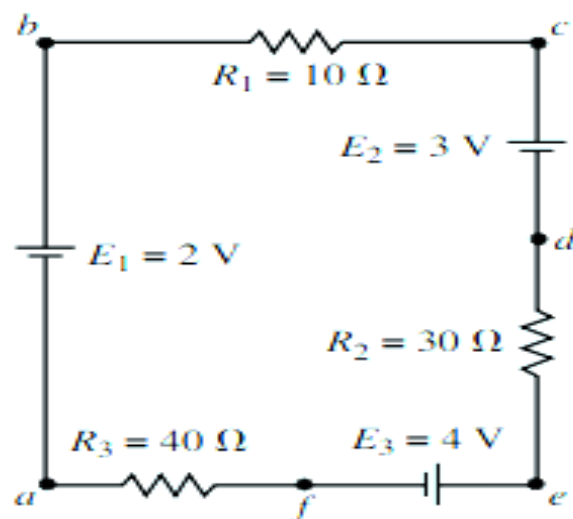


FIGURE 5-28

Solution First, we determine that the equivalent supply voltage for the circuit is

$$E_T = 3 \text{ V} + 4 \text{ V} - 2 \text{ V} = 5.0 \text{ V}$$

with a polarity such that current will move in a counterclockwise direction within the circuit.

Next, we determine the voltages on all resistors by using the voltage divider rule and assigning polarities based on the direction of the current.

$$V_1 = \frac{R_1}{R_T} E_T$$

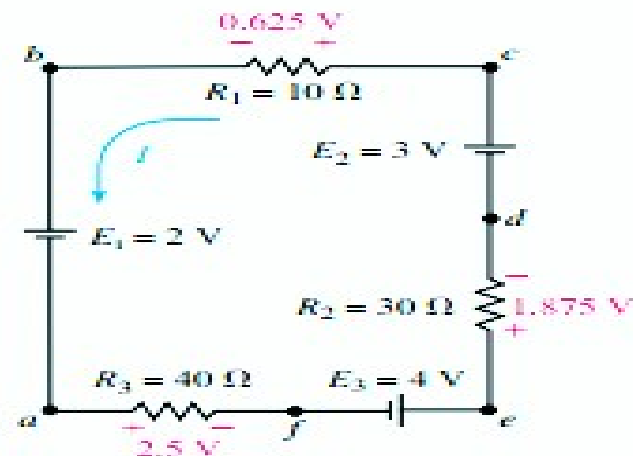
$$= \left(\frac{10 \, \Omega}{10 \, \Omega + 30 \, \Omega + 40 \, \Omega} \right) (5.0 \, \text{V}) = 0.625 \, \text{V}$$

$$V_2 = \frac{R_2}{R_T} E_T$$

$$= \left(\frac{30 \, \Omega}{10 \, \Omega + 30 \, \Omega + 40 \, \Omega} \right) (5.0 \, \text{V}) = 1.875 \, \text{V}$$

$$V_3 = \frac{R_3}{R_T} E_T$$

$$= \left(\frac{40 \, \Omega}{10 \, \Omega + 30 \, \Omega + 40 \, \Omega} \right) (5.0 \, \text{V}) = 2.50 \, \text{V}$$



nally, we solve for the voltages between the indicated points:

$$V_{ac} = -2.0 \, \text{V} - 0.625 \, \text{V} = -2.625 \, \text{V}$$

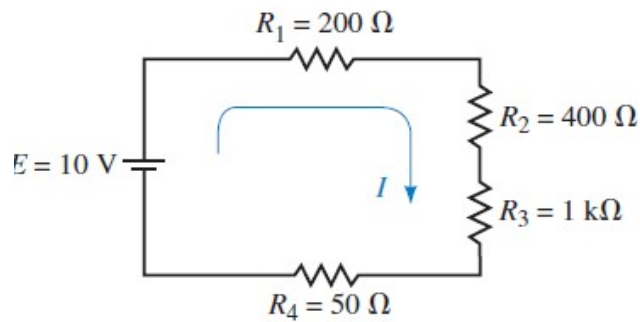
$$V_{ad} = -2.0 \, \text{V} - 0.625 \, \text{V} + 3.0 \, \text{V} = +0.375 \, \text{V}$$

$$V_{af} = +3.0 \, \text{V} - 1.875 \, \text{V} + 4.0 \, \text{V} = +5.125 \, \text{V}$$

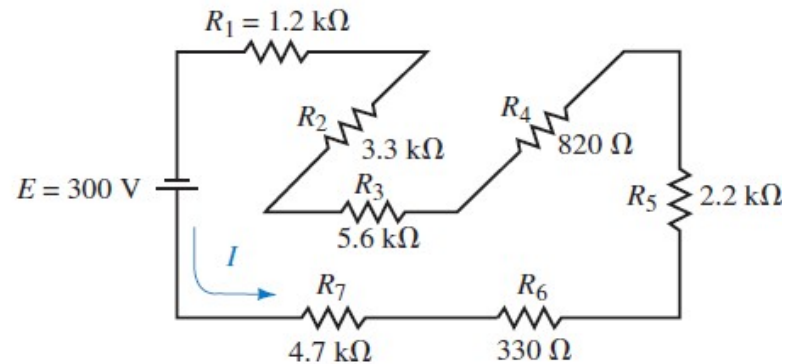
$$V_{eb} = +1.875 \, \text{V} - 3.0 \, \text{V} + 0.625 \, \text{V} = -0.500 \, \text{V}$$

Tutorials

- For the circuits shown in Fig.1, determine the total resistance, R_T , and the current, I .



(a) Circuit 1

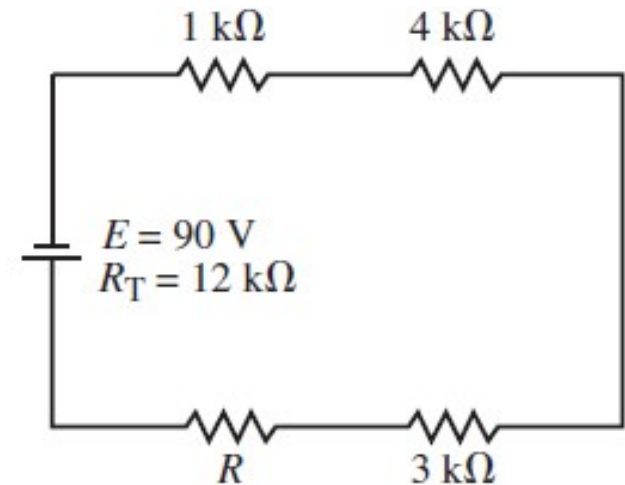


(b) Circuit 2

Fig.1

2. The circuits of Fig.2 have the total resistance, R_T , as shown. For each of the circuits find the following:

- The magnitude of current in the circuit.
- The total power delivered by the voltage source.
- The direction of current through each resistor in the circuit.
- The value of the unknown resistance, R .
- The voltage drop across each resistor.
- The power dissipated by each resistor. Verify that the summation of powers dissipated by the resistors is equal to the power delivered by the voltage source.



(a) Circuit 1

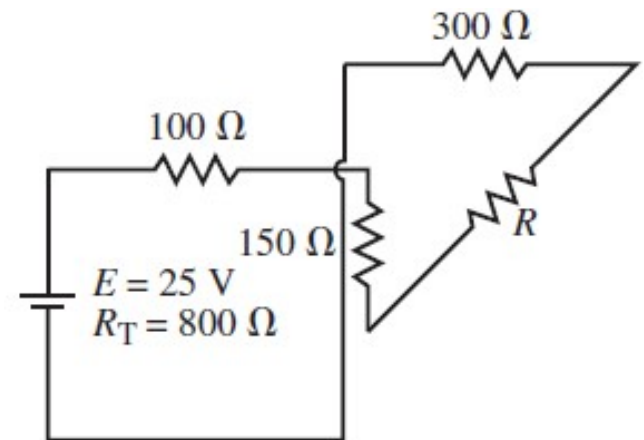


Fig.2

(b) Circuit 2

3. For the circuit of Fig.3, find the following quantities:

- The circuit current.
- The total resistance of the circuit.
- The value of the unknown resistance, R .
- The voltage drop across all resistors in the circuit.
- The power dissipated by all resistors.

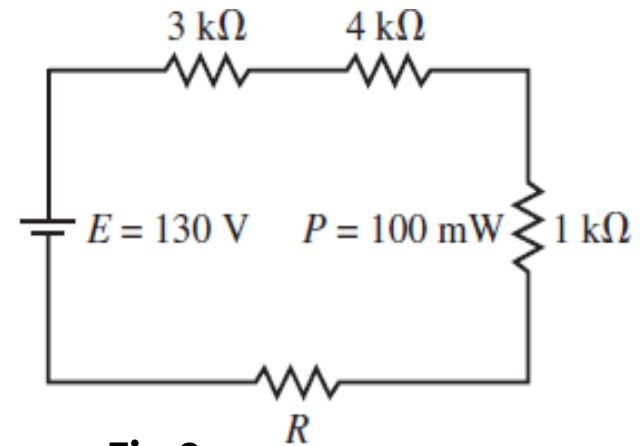


Fig.3

4. The circuit of Fig.4 has a current of 2.5 mA .

Find the following quantities:

- The total resistance of the circuit.
- The value of the unknown resistance, R_2 .
- The voltage drop across each resistor in the circuit.
- The power dissipated by each resistor in the circuit.

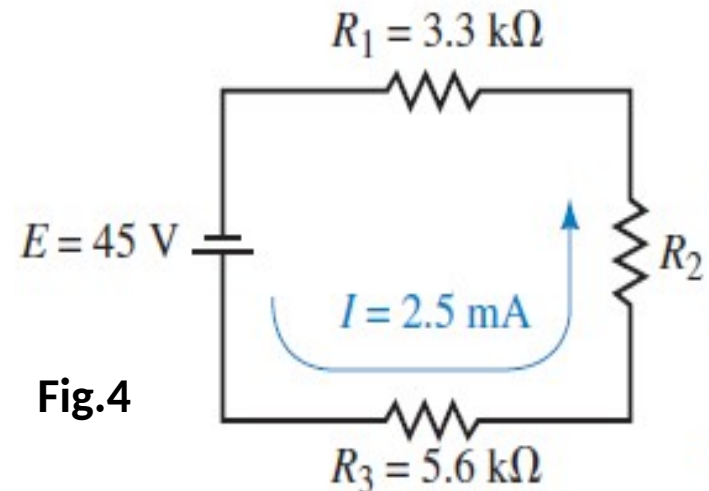
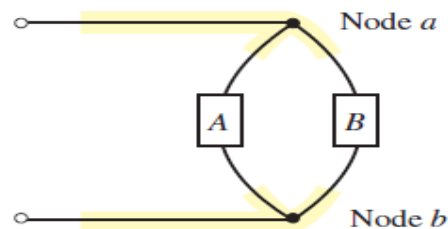


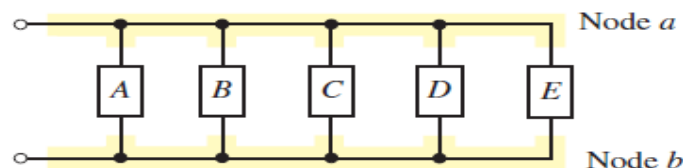
Fig.4

PARALLEL CIRCUITS

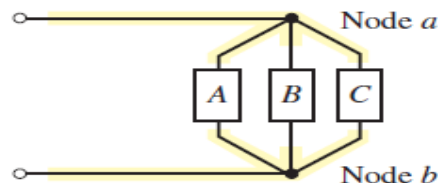
- Elements or branches are said to be in a parallel connection when **they have exactly two nodes in common**. Additionally, these parallel elements or branches **will have the same voltage across them**.
- The elements between the nodes may be any two-terminal devices such as **voltage sources, resistors, light bulbs, etc.**



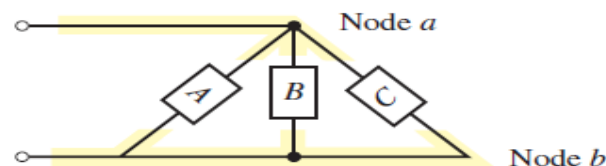
(a)



(c)



(b)



(d)

Parallel elements.

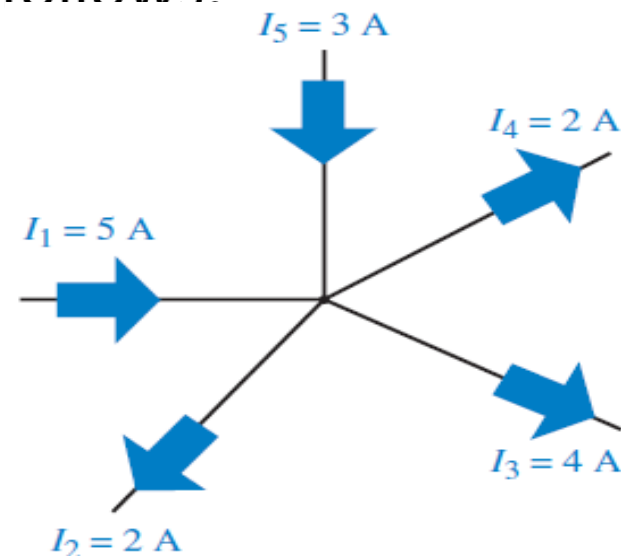
• Kirchhoff's Current Law (KCL)

- KCL is the underlying principle that is used to explain the operation of a *parallel circuit*.
- Kirchhoff's current law states the following:
 - *The summation of currents entering a node is equal to the summation of currents leaving the node.*
- In mathematical form, KCL is stated as follows:

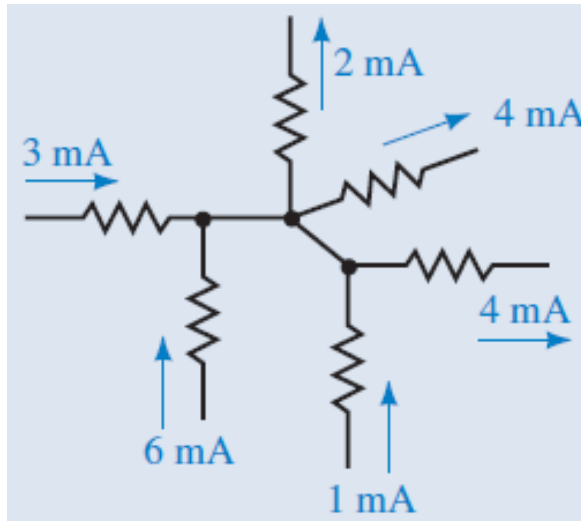
- $\sum I_{\text{entering node}} = \sum I_{\text{leaving node}}$

Example:

$$\begin{aligned}\sum I_{\text{in}} &= \sum I_{\text{out}} \\ 5 \text{ A} + 3 \text{ A} &= 2 \text{ A} + 4 \text{ A} + 8 \text{ A} \\ 8 \text{ A} &= 8 \text{ A} \text{ (checks!)}\end{aligned}$$



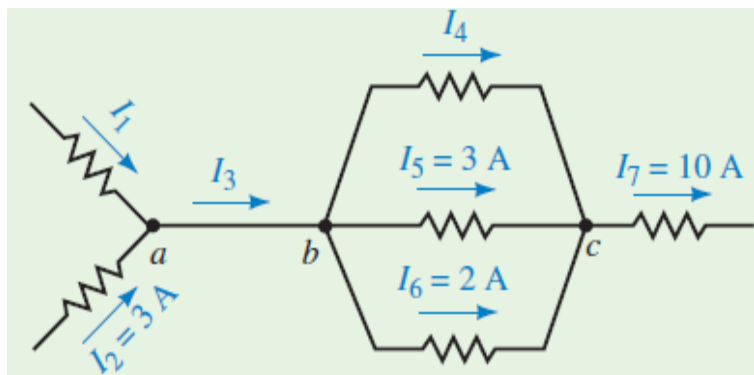
- Example1:** Verify that Kirchhoff's current law



Solution: All resistors are connected at the same node

$$3 \text{ mA} + 6 \text{ mA} + 1 \text{ mA} = 2 \text{ mA} + 4 \text{ mA} + 4 \text{ mA}$$

- Example2** Find the magnitudes of the unknown currents.



Solution: At points a and b, there are two unknown currents: At c, we have

$$I_4 + I_5 + I_6 = I_7; \equiv I_4 + 3 + 2 = 10; \equiv I_4 = 5$$

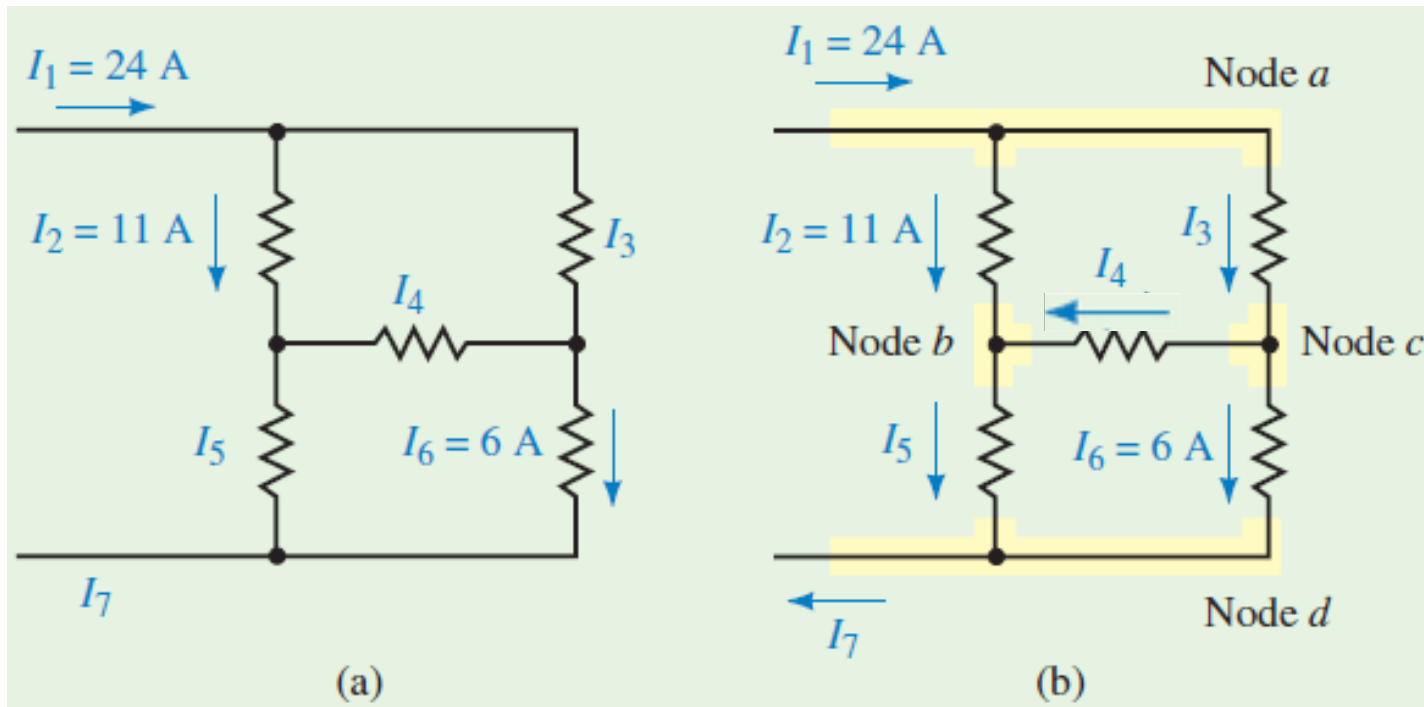
Now we can see that at point b the current entering is

$$I_3 = 5 + 3 + 2 = 10 \Omega$$

$$\text{At point a, } I_1 = 10 - 3 = 7$$

- **Example for enhancement (3)**

Determine the unknown currents in the network.

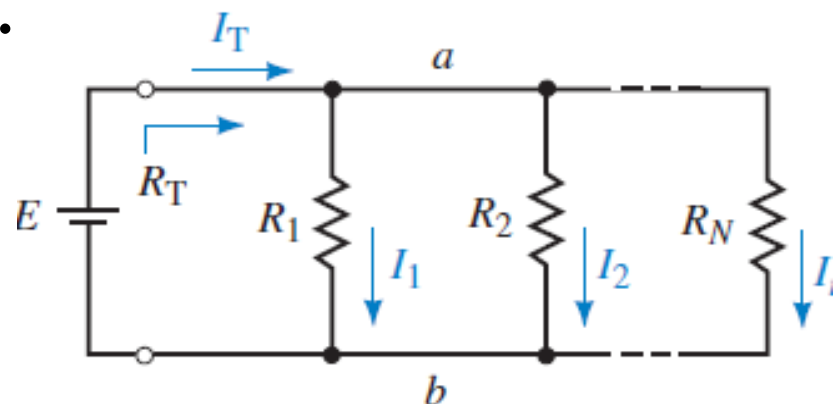


• Resistors in Parallel

- A simple parallel circuit is constructed by combining a voltage source with several resistors.
- The voltage source will result in **current from the positive terminal of the source toward node a**. At this point the current will split between the various resistors and then recombine at node **b** before continuing to the negative terminal of the voltage source.

We would find that the **voltage across all parallel resistors is exactly equal**, namely

$$V_{R1} = V_{R2} = V_{R3} = \cdots \cdots = V_{Rn} = E$$



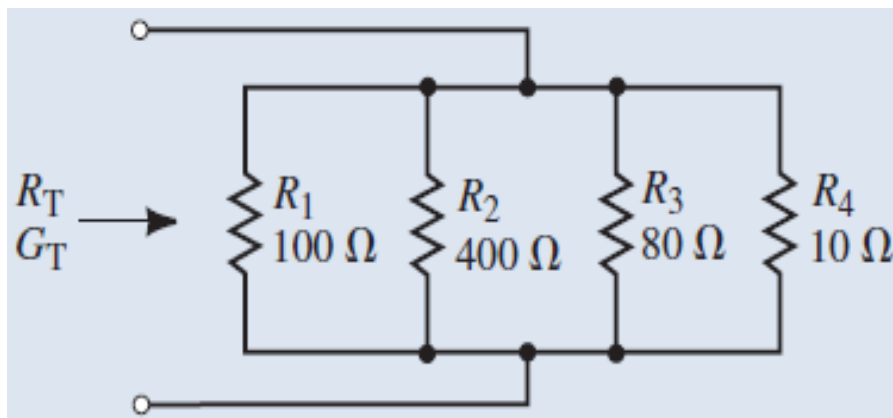
- Therefore, by applying Kirchhoff's voltage law, we make the following statement:
 - *The voltage across all parallel elements in a circuit will be the same.*
- This principle allows us to determine the equivalent resistance, R_T , of any number of resistors connected in parallel.
- The total equivalent resistance of n parallel resistors may be determined in one step as follows:

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}}$$

- *An important effect of combining parallel resistors is that the resultant resistance will always be smaller than the smallest resistor in the combination.*
- Conductance (G) is the inverse of Resistance.

$$-\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}, \quad GT = G1 + G2 + G3 + \dots + Gn \text{ (In Siemens, S)}$$

- Example:** For the parallel network of resistors shown below, find the total conductance G_T and the total resistance R_T .



Answers

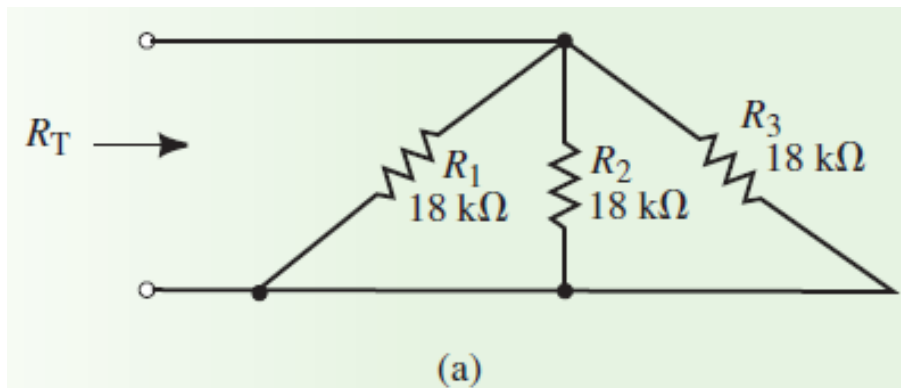
$$G_T = 0.125 \text{ S}, R_T = 8.00 \Omega$$

If we have **n equal resistors in parallel**, each resistor, R , has the same conductance G ,

The total **$G_T = nG$**

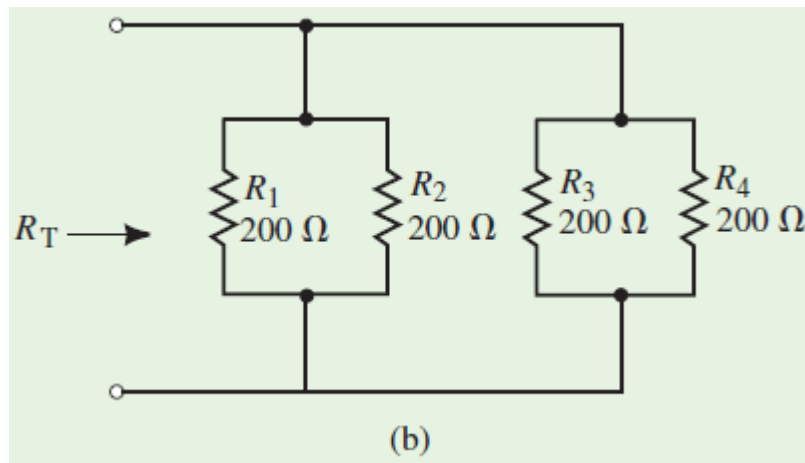
The **$R_T = 1/G_T = 1/nG = R/n$**

- Example:** For the following networks, calculate the total resistance.



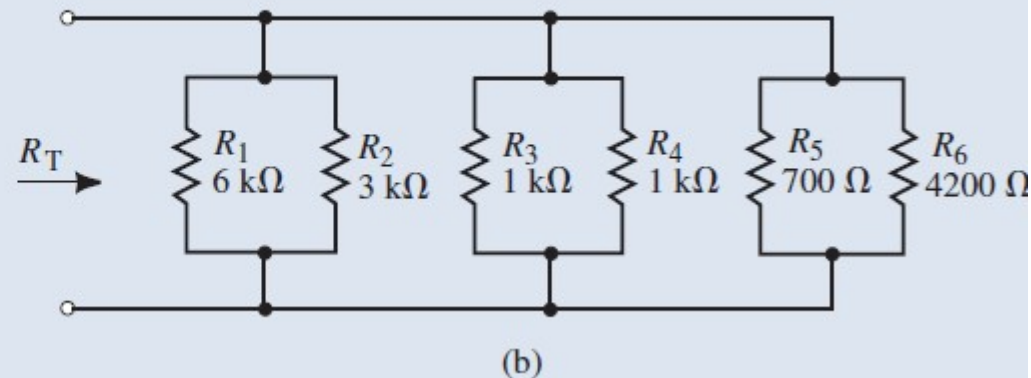
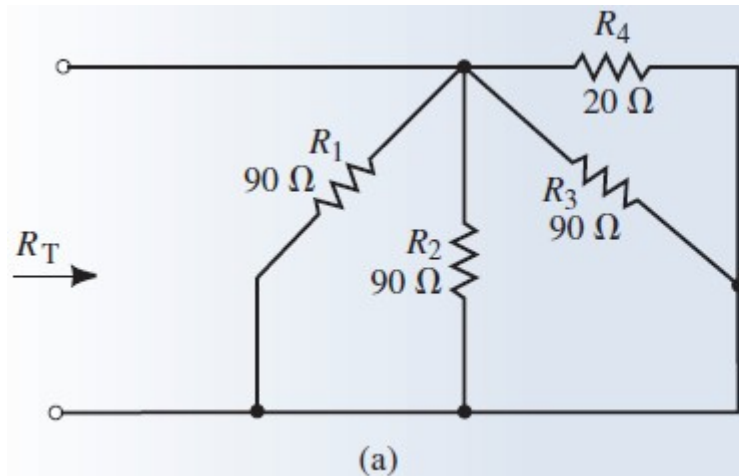
Solution

$$\text{a. } R_T = \frac{18\text{ k}\Omega}{3} = 6\text{ k}\Omega$$



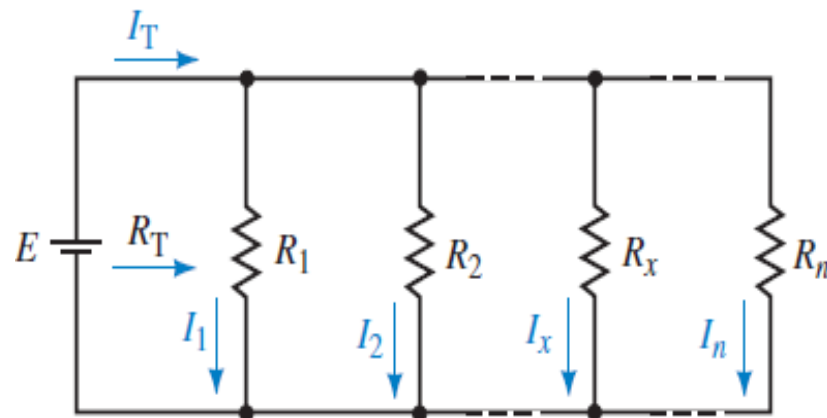
$$\text{b. } R_T = \frac{200\text{ }\Omega}{4} = 50\text{ }\Omega$$

- **Example for enhancement (4):**
- Find the total equivalent resistance for each network.



- **Current Divider Rule**
- **The voltage divider rule (VDR)** discussed earlier, was used to determine the voltage across all resistors within a **series network**.
- In **parallel networks**, the voltage **across all parallel elements** is **the same**.
- However, the **currents through the various elements** are **typically different**.
- **The current divider rule (CDR)** is used to determine how current entering a node is split between the **various parallel resistors** connected to the node.

- Total current in the circuit is $I_T = \frac{E}{R_T}$,
- Since each of the **n parallel resistors** has the same voltage, E , across its terminals, the current through any resistor in the network is given as
- $I_x = \frac{E}{R_x}$, and $E = I_T R_T$
- Hence, **$I_x = \frac{R_T}{R_x} I_T$** ,
 - Which is the current divider rule.



- If the network **consists of only two parallel resistors**, then the current through each resistor may be found in a slightly different way.

- Recall that: $R_T = \frac{R_1 \times R_2}{R_1 + R_2}$, Then the Current

- Which is simplified as:

$$I_1 = \frac{R_2}{R_1 + R_2} I_T$$

$$I_2 = \frac{R_1}{R_1 + R_2} I_T$$

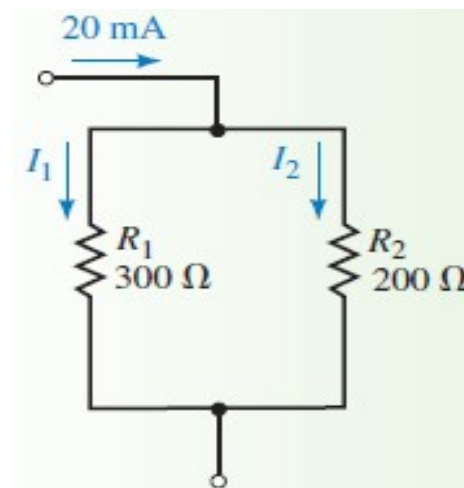
$$I_1 = \frac{I_T R_T}{R_1} = \frac{I_T \left(\frac{R_1 R_2}{R_1 + R_2} \right)}{R_1}$$

Example: Determine the currents I_1 and I_2 in the network

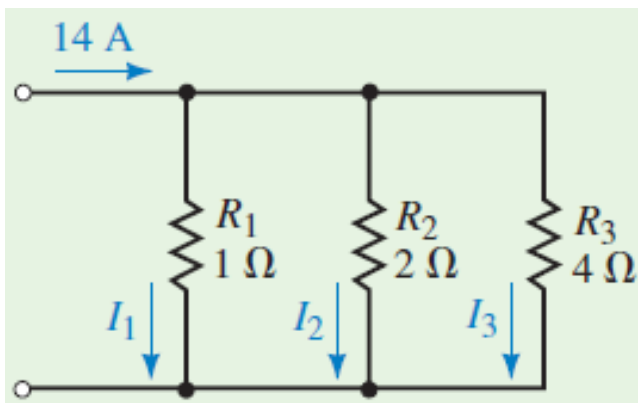
Solution: Since we have only two resistors, we have:

$$I_1 = \frac{R_2}{R_1 + R_2} I_T = \left(\frac{200 \Omega}{300 \Omega + 200 \Omega} \right) (20 \text{ mA}) = 8.00 \text{ mA}$$

$$I_2 = \frac{R_1}{R_1 + R_2} I_T = \left(\frac{300 \Omega}{300 \Omega + 200 \Omega} \right) (20 \text{ mA}) = 12.0 \text{ mA}$$



- Example:** For this network, determine the currents I_1 , I_2 , and I_3 .



Solution: We calculate the total conductance of the network.

$$G_T = \frac{1}{1 \Omega} + \frac{1}{2 \Omega} + \frac{1}{4 \Omega} = 1.75 \text{ S}$$

$$I_1 = \frac{G_1}{G_T} I_T = \left(\frac{1 \text{ S}}{1.75 \text{ S}} \right) 14 \text{ A} = 8.00 \text{ A}$$

$$I_2 = \frac{G_2}{G_T} I_T = \left(\frac{0.5 \text{ S}}{1.75 \text{ S}} \right) 14 \text{ A} = 4.00 \text{ A}$$

$$I_3 = \frac{G_3}{G_T} I_T = \left(\frac{0.25 \text{ S}}{1.75 \text{ S}} \right) 14 \text{ A} = 2.00 \text{ A}$$

Using conductance G

$$R_T = \frac{1}{G_T} = \frac{1}{1.75 \text{ S}} = 0.571 \Omega$$

$$I_1 = \frac{R_T}{R_1} I_T = \left(\frac{0.571 \Omega}{1 \Omega} \right) 14 \text{ A} = 8.00 \text{ A}$$

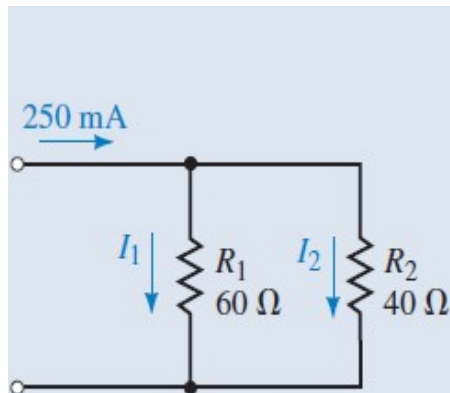
$$I_2 = \frac{R_T}{R_2} I_T = \left(\frac{0.571 \Omega}{2 \Omega} \right) 14 \text{ A} = 4.00 \text{ A}$$

$$I_3 = \frac{R_T}{R_3} I_T = \left(\frac{0.571 \Omega}{5 \Omega} \right) 14 \text{ A} = 2.00 \text{ A}$$

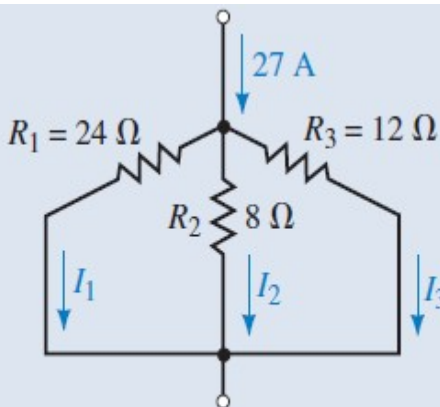
Using total resistance

- **Example for enhancement (5)**

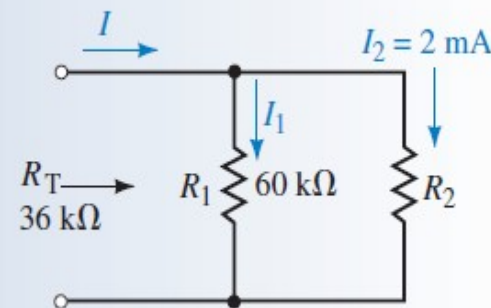
(1) Use the current divider rule to calculate the **unknown currents** for the networks below.



(a) Network 1



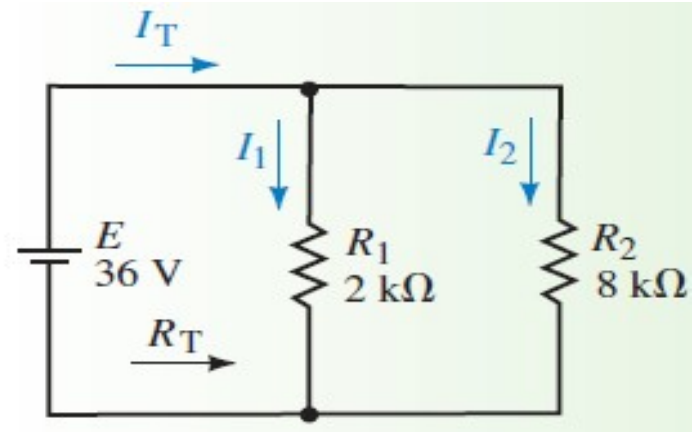
(b) Network 2



(c) Network 3

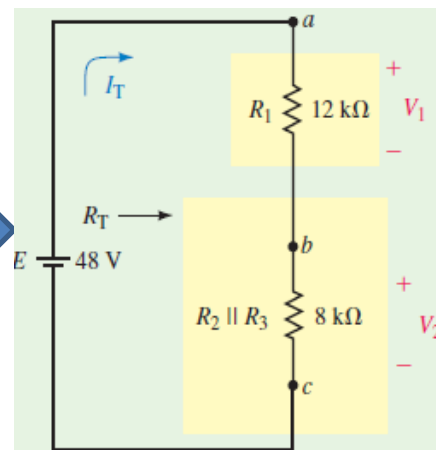
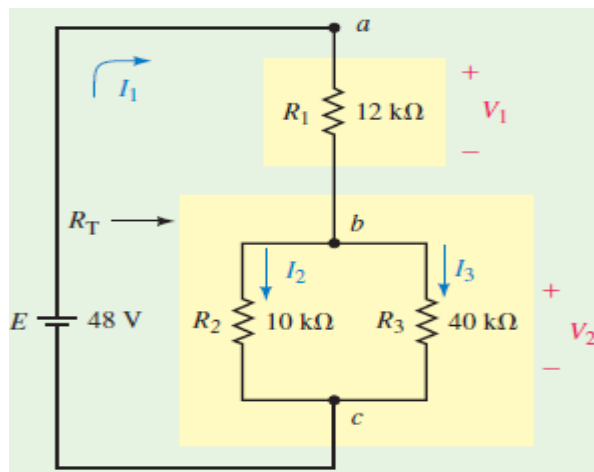
- **Example for enhancement (5)-Cont'**

- For the circuit of this figure, determine the following quantities:
- R_T , I_T
- Power delivered by the voltage source
- I_1 and I_2 using the current divider rule
- Power dissipated by the resistors.



• The Series-Parallel Network

- In analyzing such circuits, it is imperative to remember that the **rules for analyzing series and parallel** elements still apply.
 - The same current occurs through all series elements.
 - The same voltage occurs across all parallel elements.
- In addition, remember that **Kirchhoff's voltage law** and **Kirchhoff's current law** apply for all circuits regardless of whether the circuits are series, parallel, or series-parallel.
- **Example:** (a) Find R_T , (b) Calculate I_1 , I_2 , and I_3 , (c) The voltages V_1 and V_2 .



$$R_T = R_1 + R_2 || R_3$$

$$R_T = 12 \text{ k}\Omega + \frac{(10 \text{ k}\Omega)(40 \text{ k}\Omega)}{10 \text{ k}\Omega + 40 \text{ k}\Omega}$$

$$= 12 \text{ k}\Omega + 8 \text{ k}\Omega = 20 \text{ k}\Omega$$

$$I_T = I_1 = \frac{48 \text{ V}}{20 \text{ k}\Omega} = 2.4 \text{ mA}$$

$$I_2 = \frac{(40 \text{ k}\Omega)(2.4 \text{ mA})}{10 \text{ k}\Omega + 40 \text{ k}\Omega} = 1.92 \text{ mA}$$

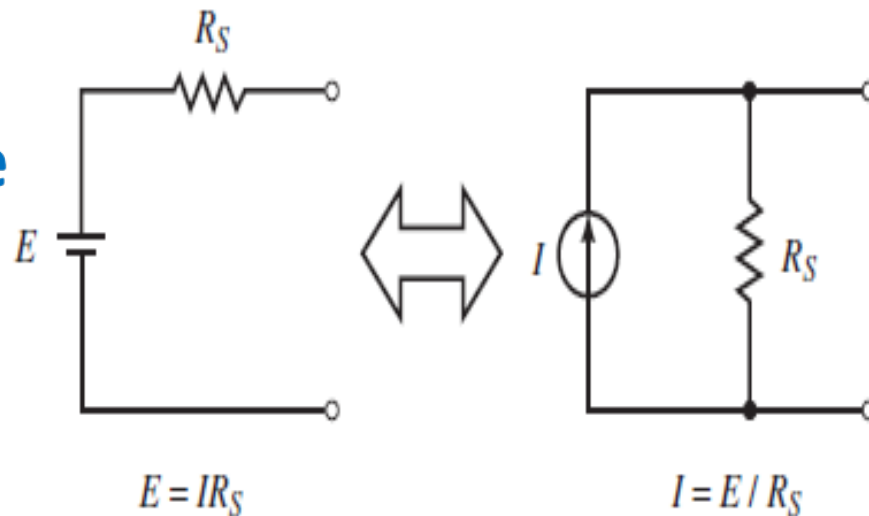
$$I_3 = \frac{(10 \text{ k}\Omega)(2.4 \text{ mA})}{10 \text{ k}\Omega + 40 \text{ k}\Omega} = 0.48 \text{ mA}$$

$$V_1 = (2.4 \text{ mA})(12 \text{ k}\Omega) = 28.8 \text{ V}$$

$$V_3 = (0.48 \text{ mA})(40 \text{ k}\Omega) = 19.2 \text{ V} = V_2$$

- **Source Transformation**
- As you recall, **voltage sources always have some series resistance**, although in some cases this resistance is **so small in comparison with other circuit resistance** that it may effectively **be ignored** when determining the operation of the circuit.
- Similarly, **a constant-current source will always have some shunt (or parallel) resistance**.
- If this resistance is very large in comparison with the other circuit resistance, **the internal resistance of the source may once again be ignored**.
- An ideal current source has an infinite shunt resistance.

- If the internal resistance of a source is considered, the source, whether it is a voltage source or a current source, is easily converted to the other type.
- In this fig. the current source is equivalent to the voltage source if $I = \frac{E}{R_S}$, and the resistances in both cases is R_S .
- Similarly, a current source may be converted to an equivalent voltage source by letting $E = IR_S$



- These results may be **easily verified** by connecting an external resistance, R_L , across each source.
- The **sources can be equivalent only if the voltage across R_L is the same** for both sources.
- Similarly, **the sources are equivalent only if the current through R_L is the same** when connected to either source.

- Therefore, Considering **Fig. (a)**, V_L and I_L are given by:

$$V_L = \frac{R_L}{R_L + R_S} E$$

$$I_L = \frac{E}{R_L + R_S}$$

- Considering **Fig. (b)**, and by converting the source we get

$$I_L = \frac{R_S}{R_S + R_L} I \quad \text{and} \quad I = \frac{E}{R_S}, \quad \text{and so we have} \quad I_L = \left(\frac{R_S}{R_S + R_L} \right) \left(\frac{E}{R_S} \right)$$

- The voltage across the resistor is given as

$$V_L = I_L R_L = \left(\frac{E}{R_S + R_L} \right) R_L$$

Which is precisely the same as the result obtained in **Fig. (a)**.

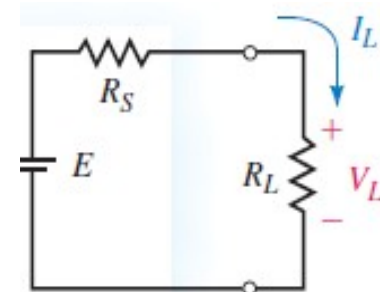


Fig.(a)

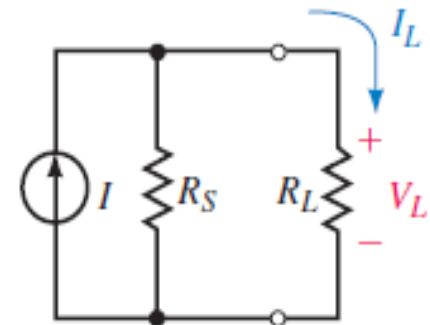
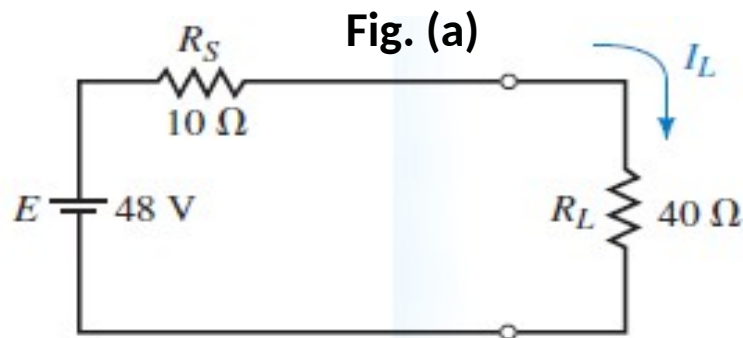
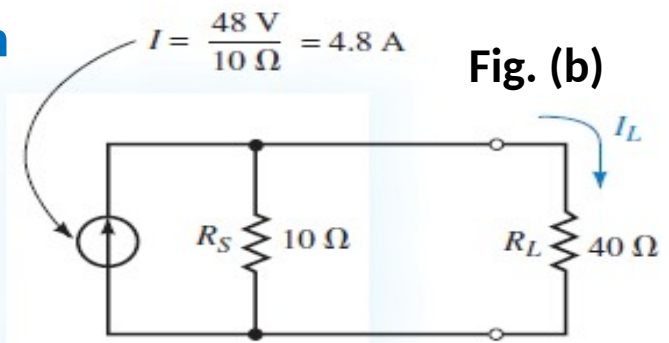


Fig.(b)

- Example 1:** Convert the voltage source of **fig. (a)** into a current source and verify that the current, I_L , through the load is the same for each source.



Solution



The equivalent current source will have a current magnitude given as: $I = \frac{48 \text{ V}}{10 \Omega} = 4.8 \text{ A}$ and the current through the load in **Fig (a)** is found as:

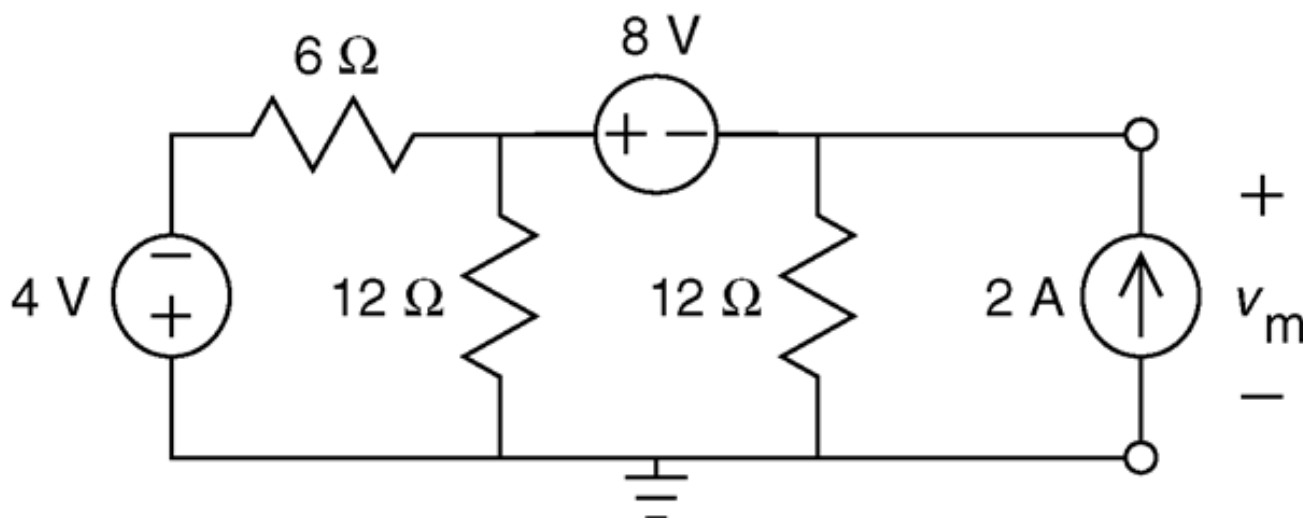
$$I_L = \frac{48 \text{ V}}{10 \Omega + 40 \Omega} = 0.96 \text{ A}$$

Also, the current through the load of Fig. (b) is: $I_L = \frac{R_S}{R_S + R_L} I_T$

$$I_L = \frac{(4.8 \text{ A})(10 \Omega)}{10 \Omega + 40 \Omega} = 0.96 \text{ A}$$

Hence, Clearly the results are the same.

- **Example for Enhancement:**
- Reduce the following circuit by using the source transformation method. Also calculate the current and the voltage dropped in the resistor(s) of the final reduced circuit



Mesh (loop) and Nodal Analysis.

Methods of Analysis to solve Branch-current ckts.

- **Main Steps:**

1. Arbitrarily *assign current directions to each branch* in the network.
2. Using the assigned currents directions, *label the polarities of the voltage drops across all resistors* in the circuit.
3. *Apply Kirchhoff's voltage* law around each of the closed loops. Write just enough equations to include all branches in the loop equations.
4. *Apply Kirchhoff's current law* at enough nodes to ensure that all branch currents have been included. In the event that a branch has only a current source, it will need to be included in this step.
5. **Solve** the resulting simultaneous **linear equations**.

- **Example** Find the current in each branch in the circuit of the following figure:

Solution

- **Step 1:** Assign **currents directions** as shown in Fig.(a).
- **Step 2:** **Indicate the polarities** of the voltage drops on all resistors in the circuit, using the assumed current directions.
- **Step 3:** Write the **Kirchhoff voltage law** equations.

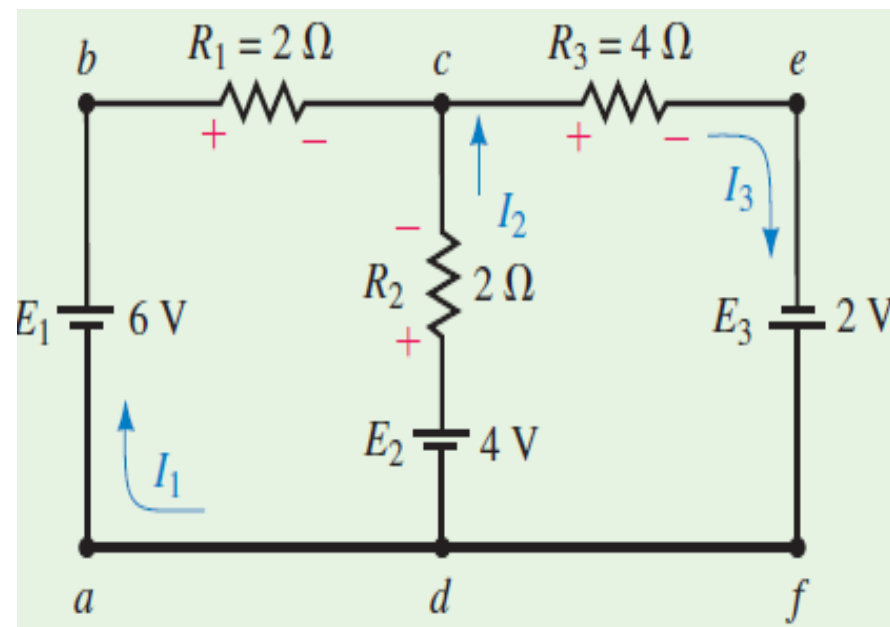


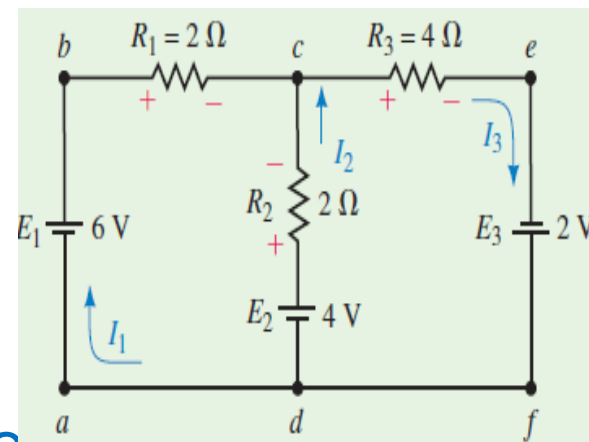
Fig. (a)

- Identify loops and Apply KVL
- We have 2 loops: Loop *abcda* and *cefdc* (or *abcefda*):

- Loop *abcda*:

$$6\text{ V} - (2\ \Omega)I_1 + (2\ \Omega)I_2 - 4\text{ V} = 0\text{ V}$$
- Loop *cefdc*:

$$4\text{ V} - (2\ \Omega)I_2 - (4\ \Omega)I_3 + 2\text{ V} = 0\text{ V}$$



- Now, all branches have been included in the loop equations, there is no need to write any more.
- **Step 4:** Write the Kirchhoff current law equation(s).

- By applying **KCL at node c**, all branch currents in the network are included.

- $I_3 = I_1 + I_2$

- To simplify the solution of the simultaneous linear equations we write them as follows:

$$2I_1 - 2I_2 + 0I_3 = 2$$

$$0I_1 - 2I_2 - 4I_3 = -6$$

$$1I_1 + 1I_2 - 1I_3 = 0$$

- By using determinant (matrix) method, we have

$$\begin{aligned} D &= \begin{vmatrix} 2 & -2 & 0 \\ 0 & -2 & -4 \\ 1 & 0 & -1 \end{vmatrix} \\ &= 2 \begin{vmatrix} -2 & -4 \\ 1 & -1 \end{vmatrix} - 0 \begin{vmatrix} -2 & 0 \\ 1 & -1 \end{vmatrix} + 1 \begin{vmatrix} -2 & 0 \\ -2 & -4 \end{vmatrix} \\ &= 2(2 + 4) - 0 + 1(8) = 20 \end{aligned}$$

By solving for the currents, we have:

$$I_1 = 1.200\text{A}$$

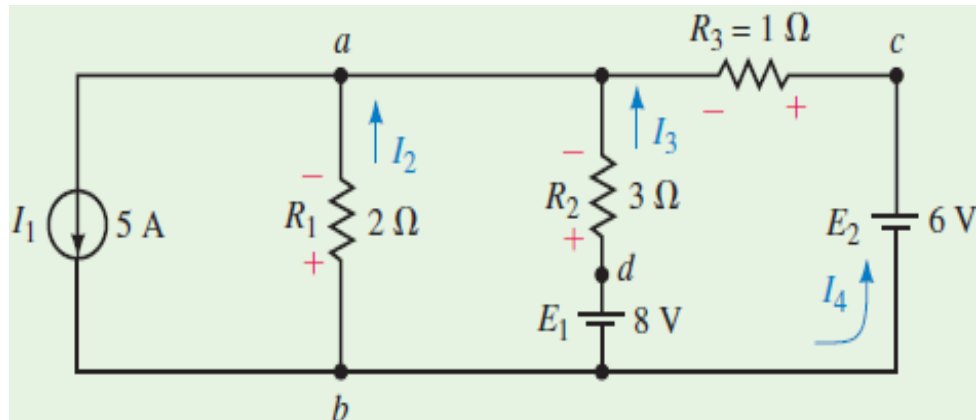
$$I_2 = 0.200\text{A}$$

$$I_3 = 1.400\text{A}$$

• Examples for Enhancement:

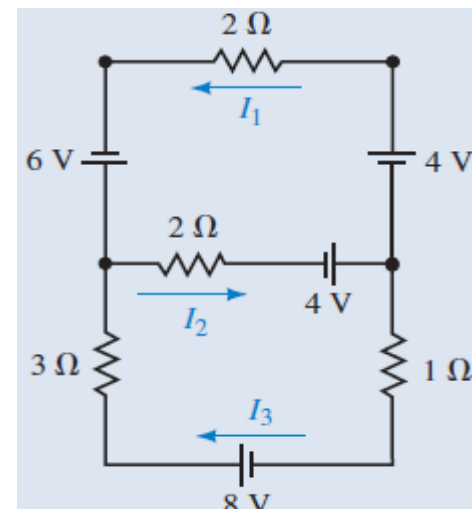
- (1) Find the currents in each branch of the circuit shown in Fig.(1).
Solve for the voltage V_{ab} .

Fig. (1)



- (2) Use branch-current analysis to solve for the indicated currents in the circuit of Fig. (2)

Fig. (2)



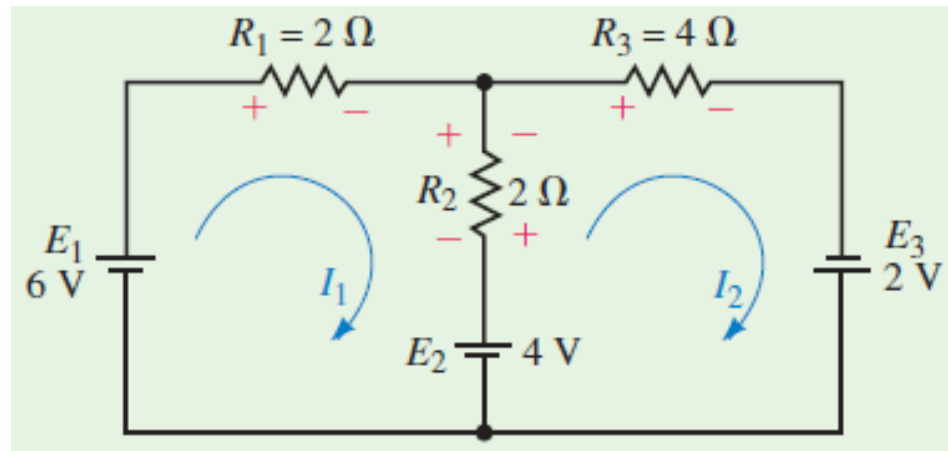
Mesh (Loop) Analysis

- The best approach to analyze the **Branch-current** circuits is to use “**mesh (or loop) analysis**”.
- The principal difference **between mesh analysis and branch-current** analysis is that we simply need to apply KVL around closed loops without the need for applying KCL.

- **Steps:**

1. **Arbitrarily assign** a clockwise current to each interior closed loop in the network. Although the assigned current may be in any direction, a clockwise direction is used to make later work simpler.
2. Using the assigned loop currents, indicate **the voltage polarities** across all resistors in the circuit. **For a resistor that is common to two loops, the polarities of the voltage drop due to each loop current should be indicated on the appropriate side of the component.**
3. Applying KVL, write the loop equations for each loop in the network. **Do not forget that resistors that are common to two loops will have two voltage drops**, one due to each loop.
4. Solve the resultant simultaneous linear equations.
5. Branch currents are determined by algebraically combining the loop currents that are common to the branch.

- **Example:**
- Find the current in each branch for the circuit below.



Solution

Step 1: Loop currents I_1 and I_2 are assigned as shown above.

Step 2: Voltage polarities are assigned according to the loop currents. Notice that the resistor R_2 has two different voltage polarities due to the different loop currents.

Step 3: The loop equations are written by applying Kirchhoff's voltage law in each of the loops. The equations are as follows:

$$\text{Loop 1: } 6\text{ V} - (2\ \Omega)I_1 - (2\ \Omega)I_1 + (2\ \Omega)I_2 - 4\text{ V} = 0$$

$$\text{Loop 2: } 4\text{ V} - (2\ \Omega)I_2 + (2\ \Omega)I_1 - (4\ \Omega)I_2 + 2\text{ V} = 0$$

- The preceding loop equations may be simplified as follows:

$$\text{Loop 1: } (4 \, \Omega)I_1 - (2 \, \Omega)I_2 = 2 \, \text{V}$$

$$\text{Loop 2: } -(2 \, \Omega)I_1 + (6 \, \Omega)I_2 = 6 \, \text{V}$$

Using determinants, the loop equations are easily solved as

$$I_1 = \frac{\begin{vmatrix} 2 & -2 \\ 6 & 6 \end{vmatrix}}{\begin{vmatrix} 4 & -2 \\ -2 & 6 \end{vmatrix}} = \frac{12 + 12}{24 - 4} = \frac{24}{20} = 1.20 \, \text{A}$$

and

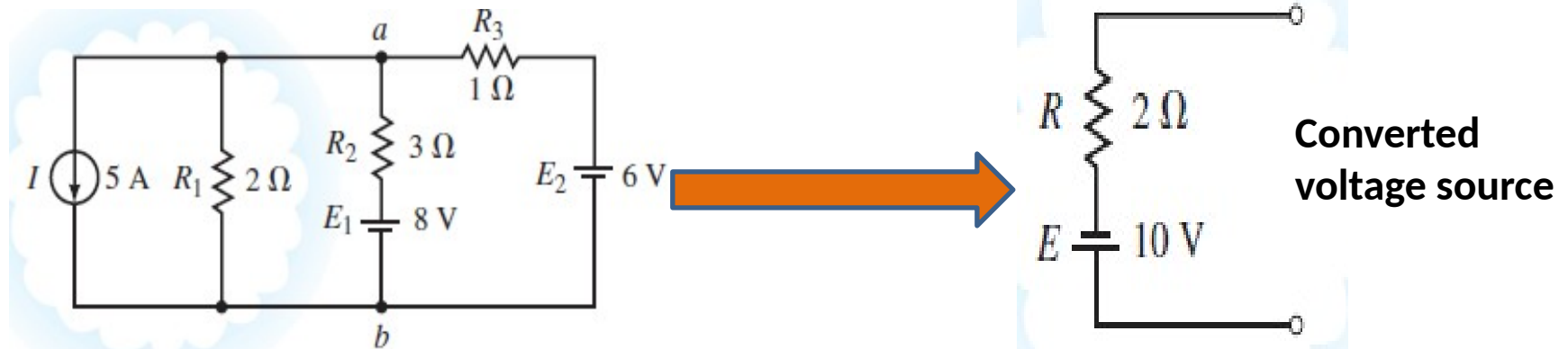
$$I_2 = \frac{\begin{vmatrix} 4 & 2 \\ -2 & 6 \end{vmatrix}}{\begin{vmatrix} 4 & -2 \\ -2 & 6 \end{vmatrix}} = \frac{24 + 4}{24 - 4} = \frac{28}{20} = 1.40 \, \text{A}$$

The branch current for **R2** is found by combining the loop currents through this resistor:

$$I_{R_2} = 1.40 \, \text{A} - 1.20 \, \text{A} = 0.20 \, \text{A (upward)}$$

- The results obtained by using mesh analysis are **exactly the same** as those obtained by branch-current analysis.
- However, **branch-current analysis required three equations**;
- While **Mesh (Loop)** approach requires the solution of only **two simultaneous** linear **equations**.
- Hence, **Mesh analysis is preferred** to branch-current analysis.
- If the **circuit being analyzed contains current sources**, the circuit may be simplified **by converting the current source(s) to voltage sources** and then solving the resulting network using the previous procedure.

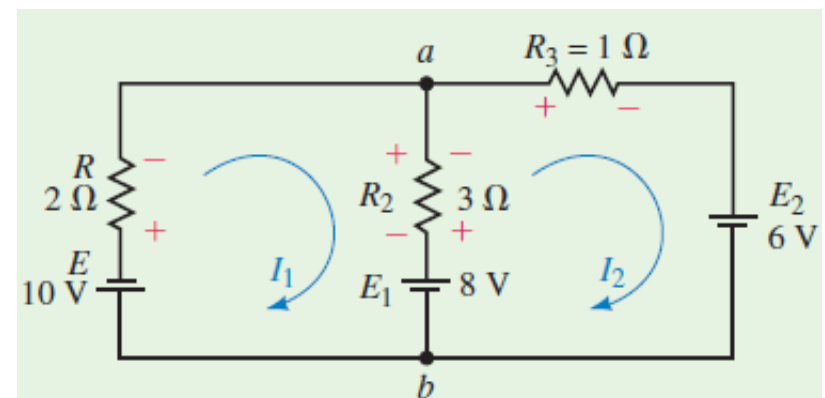
- Example:** Determine the current through the 8V battery for the following circuit.



After converting the current source into an equivalent voltage source. The equivalent circuit may now be analyzed by using the loop currents as shown below:

Solution

Only two loops



$$\text{Loop 1:} \quad -10 \text{ V} - (2 \Omega)I_1 - (3 \Omega)I_1 + (3 \Omega)I_2 - 8 \text{ V} = 0$$

$$\text{Loop 2:} \quad 8 \text{ V} - (3 \Omega)I_2 + (3 \Omega)I_1 - (1 \Omega)I_2 - 6 \text{ V} = 0$$

Rewriting the linear equations, you get the following:

$$\text{Loop 1:} \quad (5 \Omega)I_1 - (3 \Omega)I_2 = -18 \text{ V}$$

$$\text{Loop 2:} \quad -(3 \Omega)I_1 + (4 \Omega)I_2 = 2 \text{ V}$$

Solving the equations using determinants, we have the following:

$$I_1 = \frac{\begin{vmatrix} -18 & -3 \\ 2 & 4 \end{vmatrix}}{\begin{vmatrix} 5 & -3 \\ -3 & 4 \end{vmatrix}} = -\frac{66}{11} = -6.00 \text{ A}$$

$$I_2 = \frac{\begin{vmatrix} 5 & -18 \\ -3 & 2 \end{vmatrix}}{11} = -\frac{44}{11} = -4.00 \text{ A}$$

If the assumed direction of current in the 8-V battery is taken to be I_2 , then

$$I = I_2 - I_1 = -4.00 \text{ A} - (-6.00 \text{ A}) = 2.00 \text{ A}$$

The direction of the resultant current is the same as I_2 (upward).

- Example for Enhancement.**

(1) Solve for the currents through R_2 and R_3 in the fig. (1) circuit

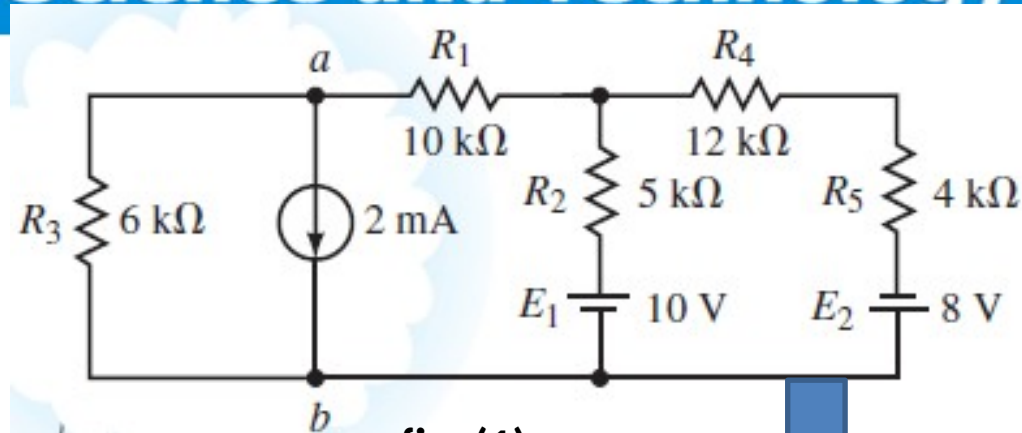


fig. (1)

Hint: Converted voltage source

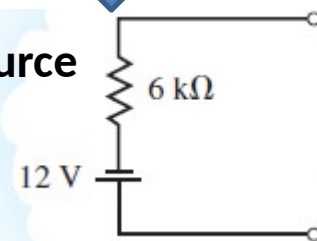
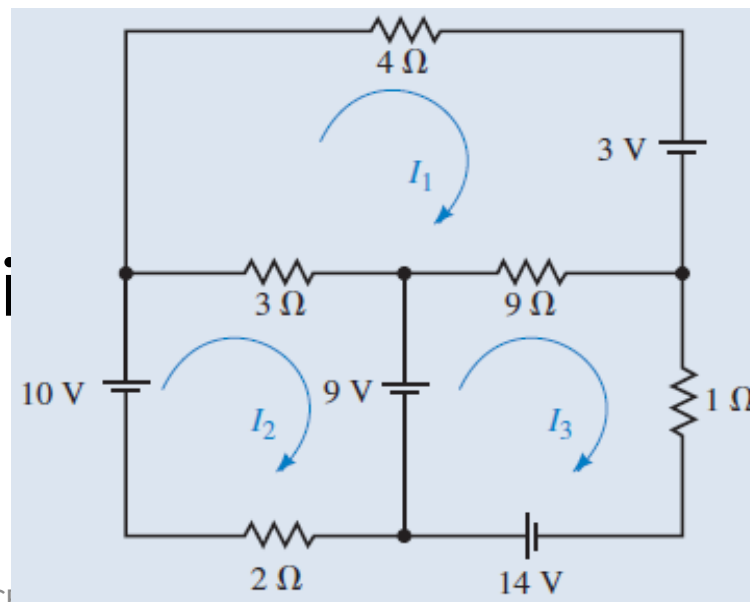


Fig. (2).

(2) Use mesh analysis to find the loop currents in Fig. (2).

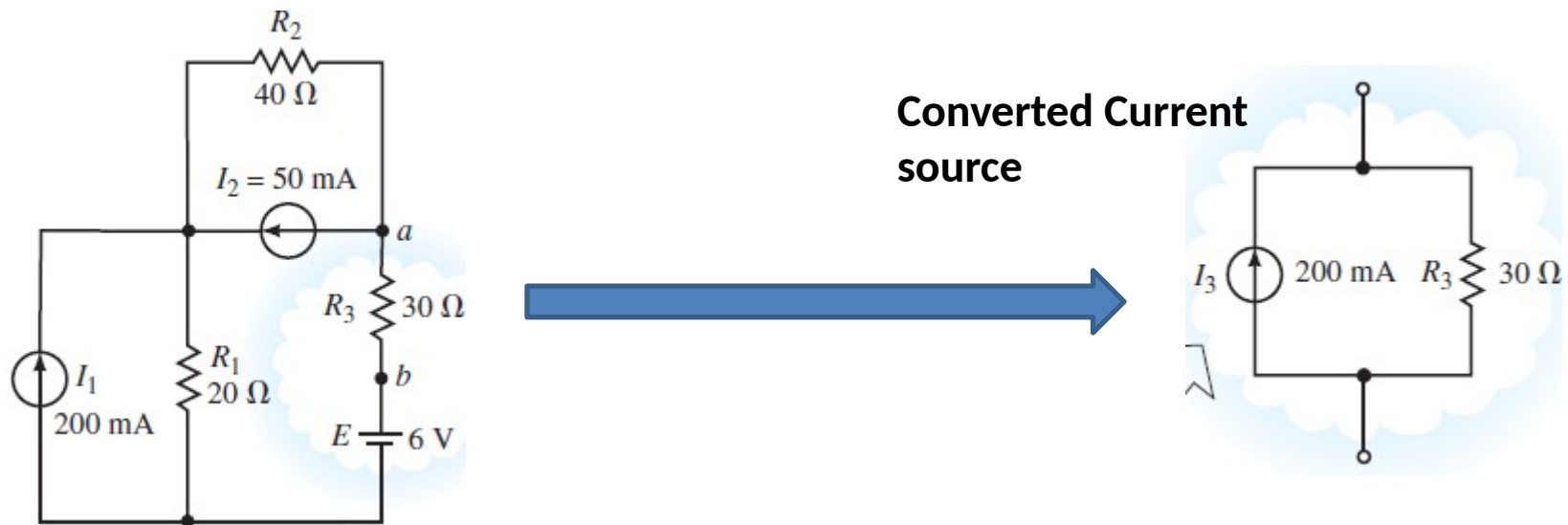


Nodal Analysis

- In Nodal analysis, we will apply **KCL to** determine the potential difference (voltage) at any node with respect to some **arbitrary reference point** in a network.
- **The steps to follow:**
 1. Arbitrarily assign a **reference node** within the circuit and indicate this node as ground.
 2. Convert each **voltage source in the network to its equivalent current source**. This step, although not absolutely necessary, **makes further calculations easier** to understand.

3. Arbitrarily assign voltages ($V_1, V_2, \dots, V_n$) to the remaining nodes in the circuit.
4. Arbitrarily **assign a current direction** to each branch in which there is no current source and indicate the corresponding polarities of the voltage drops on all resistors.
5. With the exception of ground, **apply KCL at each of the nodes**. If a circuit has a total of **$n+1$** nodes (including the reference node), there will be **n** simultaneous linear equations.
6. **Rewrite** each of the arbitrarily assigned currents in terms of the potential difference across a known resistance.
7. **Solve** the resulting simultaneous linear equations for the voltages ($V_1, V_2, \dots, V_n$).

- Example:** Given this circuit, use nodal analysis to solve for the voltage V_{ab} .



- $I = V/R = 6/30 = 0.2\text{ A}$ or 200 mA

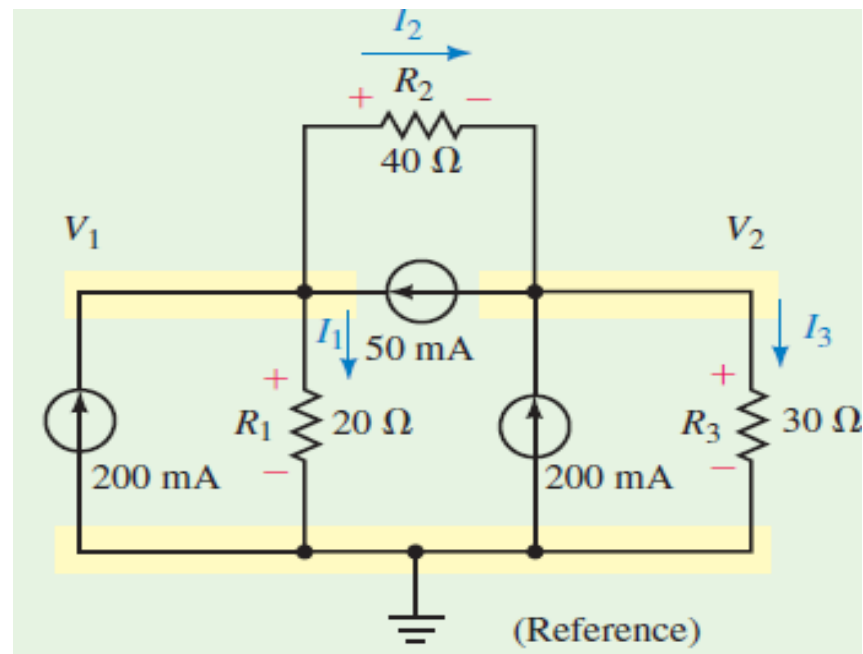
• Solution

Step 1: Select a convenient reference node (ground).

Step 2: Convert the voltage sources into equivalent current sources. The equivalent circuit is shown on the right side.

Steps 3 and 4: Arbitrarily assign node voltages and branch currents. Indicate the voltage polarities across all resistors according to the assumed current directions.

Step 5: We now apply Kirchhoff's current law at the nodes labelled as V_1 and



Node V_1 :

$$\sum I_{\text{entering}} = \sum I_{\text{leaving}}$$

$$200 \text{ mA} + 50 \text{ mA} = I_1 + I_2$$

Node V_2 :

$$\sum I_{\text{entering}} = \sum I_{\text{leaving}}$$

$$200 \text{ mA} + I_2 = 50 \text{ mA} + I_3$$

- Using *nodal method*, the currents are rewritten in terms of the voltages across the resistors as follows:

$$\begin{aligned} I_1 &= \frac{V_1}{20 \, \Omega} \\ I_2 &= \frac{V_1 - V_2}{40 \, \Omega} \\ I_3 &= \frac{V_2}{30 \, \Omega} \end{aligned}$$

- The nodal equations become:

$$\begin{aligned} 200 \, \text{mA} + 50 \, \text{mA} &= \frac{V_1}{20 \, \Omega} + \frac{V_1 - V_2}{40 \, \Omega} \\ 200 \, \text{mA} + \frac{V_1 - V_2}{40 \, \Omega} &= 50 \, \text{mA} + \frac{V_2}{30 \, \Omega} \end{aligned}$$

- Substituting the voltage expressions into the original nodal equations, we have:

$$\begin{aligned} \left(\frac{1}{20 \, \Omega} + \frac{1}{40 \, \Omega} \right) V_1 - \left(\frac{1}{40 \, \Omega} \right) V_2 &= 0.25 \, \text{A} \\ -\left(\frac{1}{40 \, \Omega} \right) V_1 + \left(\frac{1}{30 \, \Omega} + \frac{1}{40 \, \Omega} \right) V_2 &= 0.15 \, \text{A} \end{aligned}$$

- Hence, solving simultaneously, the voltage V_2 is the same as the voltage V_a , namely:

$$V_a = 4.67 \, \text{V} = 6.0 \, \text{V} + V_{ab}$$

- Therefore, the voltage **V_{ab}** is simply found as:

$$V_{ab} = 4.67 \, \text{V} - 6.0 \, \text{V} = -1.33 \, \text{V}$$

These may be further simplified as

$$(0.075 \text{ S})V_1 - (0.025 \text{ S})V_2 = 250 \text{ mA}$$

$$-(0.025 \text{ S})V_1 + (0.058\bar{3})V_2 = 150 \text{ mA}$$

Step 7: Use determinants to solve for the nodal voltages as

$$\begin{aligned} V_1 &= \frac{\begin{vmatrix} 0.250 & -0.025 \\ 0.150 & 0.058\bar{3} \end{vmatrix}}{\begin{vmatrix} 0.075 & -0.025 \\ 0.025 & 0.058\bar{3} \end{vmatrix}} \\ &= \frac{(0.250)(0.058\bar{3}) - (0.150)(-0.025)}{(0.075)(0.058\bar{3}) - (-0.025)(-0.025)} \\ &= \frac{0.018\bar{3}}{0.00375} = 4.89 \text{ V} \end{aligned}$$

and

$$\begin{aligned} V_2 &= \frac{\begin{vmatrix} 0.075 & 0.250 \\ -0.025 & 0.150 \end{vmatrix}}{\begin{vmatrix} 0.075 & 0.025 \\ -0.025 & 0.058\bar{3} \end{vmatrix}} \\ &= \frac{(0.075)(0.150) - (-0.025)(0.250)}{0.00375} \\ &= \frac{0.0175}{0.00375} = 4.67 \text{ V} \end{aligned}$$

If we go back to the original circuit of Figure 8–30, we see that the voltage V_2 is the same as the voltage V_a , namely

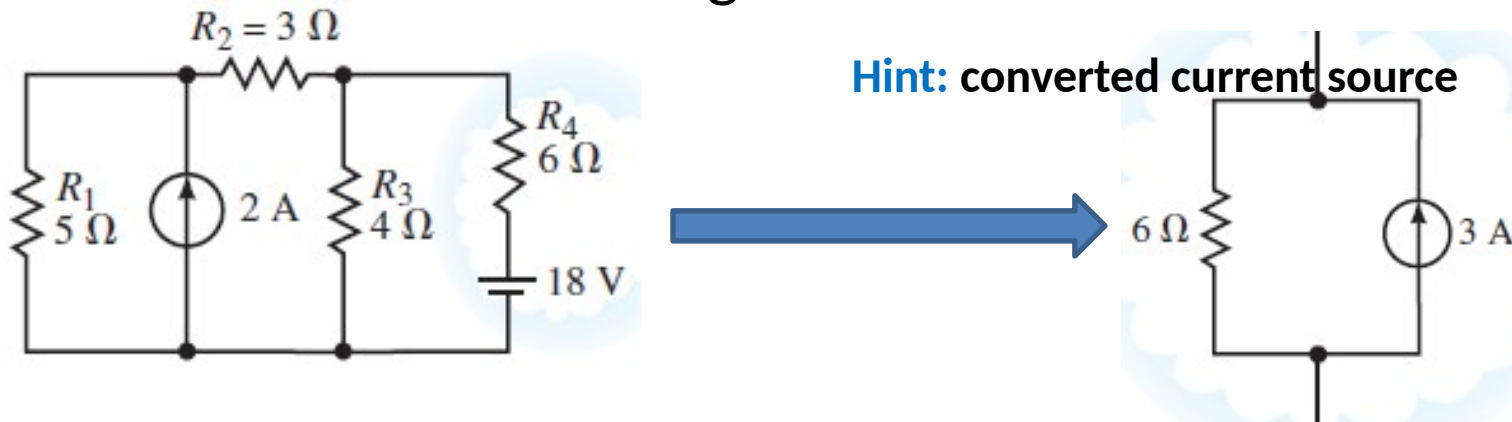
$$V_a = 4.67 \text{ V} = 6.0 \text{ V} + V_{ab}$$

Therefore, the voltage V_{ab} is simply found as

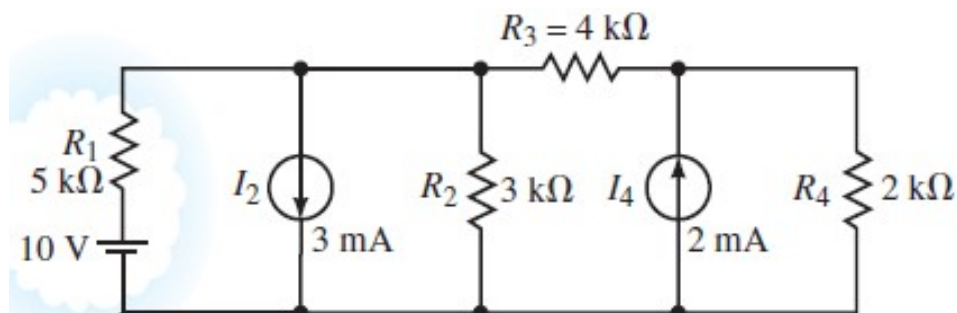
$$V_{ab} = 4.67 \text{ V} - 6.0 \text{ V} = -1.33 \text{ V}$$

- Example for Enhancement:**

(1) Determine the nodal voltages for the circuit shown.



(2) Use nodal analysis to find the nodal voltages for this circuit. Use the answers to solve for the current through R_1 .



- Assignment 1
- With atleast 2 circuit examples explain the nodal mesh (loop)analysis in circuit analysis

Next Class: **Network Theorems**